

MSCF Probability: Problem Session #1

1. Since $P(B) > 0$,

$$\begin{aligned} 1 - P(A | B) &= 1 - P(A \cap B) / P(B) \\ &= (P(B) - P(A \cap B)) / P(B) \\ &= (P(A^c \cap B)) / P(B) \\ &= P(A^c | B). \end{aligned}$$

2. (a) We are given that $P(A \cap B^c) + P(A^c \cap B) = .11$.

Let $m = P(A \cap B)$, so, $P(A \cup B) = .11 + m$, and also

$$P(A \cup B) = P(A) + P(B) - m = .85 + .9 - m$$

which implies that $.11 + m = 1.75 - m$ and hence $m = .82$. So $P(A \cup B) = .11 + .82 = .93$.

$$(b) P(A | B^c) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A) - P(A \cap B)}{P(B^c)} = \frac{.85 - .82}{1 - .9} = .3$$

$$(c) \text{ No, since, for example, } P(A \cap B) = .82 \text{ but } P(A) \cdot P(B) = .765$$

3. Define

$$\begin{aligned} A &= \{\text{student answered the question correctly.}\} \\ B &= \{\text{student actually knows the answer}\} \end{aligned}$$

We are given that $P(B) = p$ and that $P(A | B^c) = \frac{1}{m}$. Also, clearly $P(A | B) = 1$.

We need to find $P(B | A)$. Using Bayes Rule,

$$P(B | A) = \frac{P(A | B)P(B)}{P(A | B)P(B) + P(A | B^c)P(B^c)} = \frac{1 \cdot p}{1 \cdot p + \frac{1}{m}(1 - p)} = \frac{mp}{mp + (1 - p)}.$$

- 4.

$$\begin{aligned} P(A_2 | A_1 \cap B) &= \frac{P(A_2 \cap A_1 \cap B)}{P(A_1 \cap B)} \\ &= \frac{P(A_1 \cap A_2 | B)P(B)}{P(A_1 | B)P(B)} \\ &= \frac{P(A_1 | B)P(A_2 | B)P(B)}{P(A_1 | B)P(B)} \\ &\quad (\text{Since } A_1 \text{ and } A_2 \text{ are independent conditional on } B) \\ &= P(A_2 | B) \end{aligned}$$

5. Let n and m be nonnegative integers.

$$P(X \geq n + m | X \geq n) = \frac{P(X \geq n + m, X \geq n)}{P(X \geq n)} = \frac{P(X \geq n + m)}{P(X \geq n)} = \otimes$$

Note that if $X \sim \text{Geometric}(p)$, then:

$$P(X \geq k) = \sum_{x=k}^{\infty} p(1-p)^x = p \sum_{x=k}^{\infty} (1-p)^x = p \cdot \frac{(1-p)^k}{1-(1-p)} = (1-p)^k$$

So,

$$\otimes = \frac{P(X \geq n+m)}{P(X \geq n)} = \frac{(1-p)^{n+m}}{(1-p)^n} = (1-p)^m = P(X \geq m)$$

6. If X is the number of tosses prior to first “Heads,” then X is $\text{Geometric}(p)$.

$$\begin{aligned} P(\text{First heads on odd toss}) &= P(X \text{ is even}) \\ &= P(X=0) + P(X=2) + P(X=4) + \dots \\ &= p + p(1-p)^2 + p(1-p)^4 + \dots = p \sum_{k=0}^{\infty} (1-p)^{2k} \\ &= p \sum_{k=0}^{\infty} \left[(1-p)^2 \right]^k = p \cdot \frac{1}{1-(1-p)^2} \\ &= \frac{p}{p(2-p)} = \frac{1}{2-p} \end{aligned}$$

$$\text{Now, } \frac{1}{2-p} > \frac{1}{2-p+p} = \frac{1}{2}.$$

7. Let X be the number of bacteria in a 10 ml sample. We are given that $X \sim \text{Poisson}(\lambda = 1.4)$.

Let the random variable Y be the number of samples out of 20 that are “successes” which in our case means contaminated.

$$P(\text{success}) = P(X \geq 2) = 1 - P(X \leq 1) = 1 - (e^{-1.4} + 1.4e^{-1.4}) = 1 - 2.4e^{-1.4} = 0.41.$$

This implies that Y has the $\text{binomial}(n=20, p=0.41)$ distribution and that

$$P(Y=10) = \binom{20}{10} (0.41)^{10} (0.59)^{10} = 0.125$$

8. For $x \geq 0$, $F_X(x) = 1 - e^{-2x} (1 + 2x + 2x^2)$.

This implies that

$$f_X(x) = \frac{d}{dx} F_X(x) = 2e^{-2x} (1 + 2x + 2x^2) - e^{-2x} (2 + 4x) = 4x^2 e^{-2x}, \quad x \geq 0$$

Comment: We see that $X \sim \text{Gamma}(3,2)$.

9. Let time zero correspond to the time of an “event” under the combined process, i.e., the event at time zero could be of either type. Then, let X be the time of the next event of type 1, and let Y be the time of the next event of type 2. Of course, $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$, and since the Poisson processes were assumed independent, X and Y are independent. Let $T = \min(X, Y)$, so that T is the time of the next event under the combined process. From a previous exercise, we know that T has the $\text{Exp}(\lambda_1 + \lambda_2)$ distribution. Hence, the rate for the combined process is $\lambda_1 + \lambda_2$.

10. We note the following:

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$$\begin{aligned}
 E(XY) &= \int_{y=0}^{\infty} \int_{x=0}^y xy \frac{2e^{-2y}}{y} dx dy \\
 &= \int_{y=0}^{\infty} 2e^{-2y} \frac{y^2}{2} dy \\
 &= \int_{y=0}^{\infty} e^{-2y} y^2 dy, \text{ which, using Useful Math Facts,} \\
 &= \frac{\Gamma(3)}{2^3} = \frac{2!}{8} = \frac{1}{4}
 \end{aligned}$$

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$$\begin{aligned}
 f_Y(y) &= \int_{x=0}^y \frac{2e^{-2y}}{y} dx = 2e^{-2y} \quad y > 0 \\
 &\Rightarrow Y \sim \text{Exp}(2) \Rightarrow E(Y) = \frac{1}{2}
 \end{aligned}$$

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$$\begin{aligned}
 E(X) &= (\text{use guidance}) = \int_{y=0}^{\infty} \int_{x=0}^y x \frac{2e^{-2y}}{y} dx dy \\
 &= \int_{y=0}^{\infty} \frac{2e^{-2y}}{y} \frac{y^2}{2} dy = \int_{y=0}^{\infty} ye^{-2y} dy, \text{ which, using Useful Math Facts,} \\
 &= \frac{\Gamma(2)}{2^2} = \frac{1}{4}
 \end{aligned}$$

Combining the three bullets gives

$$Cov(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{4} - \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}.$$

11. Since X and Y are independent,

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a)

$$\begin{aligned}
 P(\underbrace{Y < tX}_A) &= \underbrace{\int \int_A f_{X,Y}(x, y) dx dy}_A \\
 &= \int_{y=0}^{\infty} \int_{x=y/t}^{\infty} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y} dx dy \\
 &= \int_{y=0}^{\infty} \lambda_2 e^{-\lambda_2 y} \int_{x=y/t}^{\infty} \lambda_1 e^{-\lambda_1 x} dx dy \\
 &= \lambda_2 \int_{y=0}^{\infty} e^{-\lambda_2 y} \cdot e^{-\lambda_1 y/t} dy \\
 &= \lambda_2 \int_{y=0}^{\infty} e^{-(\lambda_1/t + \lambda_2)y} dy \\
 &= \lambda_2 \left(\frac{1}{\lambda_1/t + \lambda_2} \right) = \frac{t\lambda_2}{\lambda_1 + t\lambda_2}
 \end{aligned}$$

where, on the last line, the first equality is because $\int_0^\infty e^{-ax} dx = \frac{1}{a}$ for any a (see Useful Math Facts.)

(b)

$$\begin{aligned} F_T(t) &= P(T \leq t) = P\left(\frac{Y}{X} \leq t\right) = P(Y \leq tX) \\ &= \text{(part (a) with } \lambda_1 = \lambda_2 = 1) = \frac{t}{1+t}, t > 0 \\ \Rightarrow f_T(t) &= \frac{\partial}{\partial t} F_T(t) = \frac{\partial}{\partial t} \left(\frac{t}{1+t}\right) = \frac{1}{(1+t)^2}, t > 0 \end{aligned}$$